



Universidade do Minho
Departamento de Informática

Dados e Aprendizagem Automática
Ano letivo 2025/2026

Practical Exercise no. 12

Theme Reinforcement Learning: Q-Learning vs. SARSA

Exercise Reinforcement Learning is an automatic learning paradigm that enables an agent to learn through trial and error in an interactive environment. It uses feedback from its actions and experiences to determine the next step. Within this paradigm, there are multiple techniques, including Q-learning and SARSA.

The aim of this practical exercise is to complete a series of tasks to improve your understanding of the Q-learning and SARSA models. For this purpose, the “[CartPole-v1](#)” challenge provided by OpenAI's Gym library will be used, an example of which is shown in Fig. 1. The objective is to train an intelligent bot using the aforementioned techniques to successfully complete the challenge.

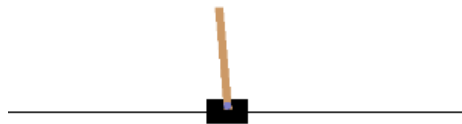


Fig. 1: The “CartPole-v1” challenge available in the OpenAI's Gym library.

Tasks This practical exercise includes the following tasks:

- T1.** Design two reinforcement learning models, trained with the respective techniques, by following the steps presented in the lab notes: (1) Q-learning and (2) SARSA.
- T2.** Critically analyse and compare the results presented by each model in terms of their average reward for each episode in the learning process and draw conclusions.
- T3.** Vary the hyperparameters of the models, such as the number of episodes, the discount factor, the learning rate, and the exploration constant. Analyse how each of these hyperparameters influences the convergence process of each model and draw conclusions.